ME530

Winter 2026

Damped Pendulum Clock

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1. Introduction

This report presents the modeling and simulation of a mechanical system in the software 20-Sim. The system is a swinging damped pendulum. In the first part the system is autonomous (simple pendulum swinging with friction), and in the second part the system is a 'pendulum clock' where impulses are applied to the pendulum each time it swings past the vertical hanging position. The impulses make up for energy lost to damping and keep the pendulum swinging indefinitely, which is the strategy used in grandfather and other mechanical clocks.

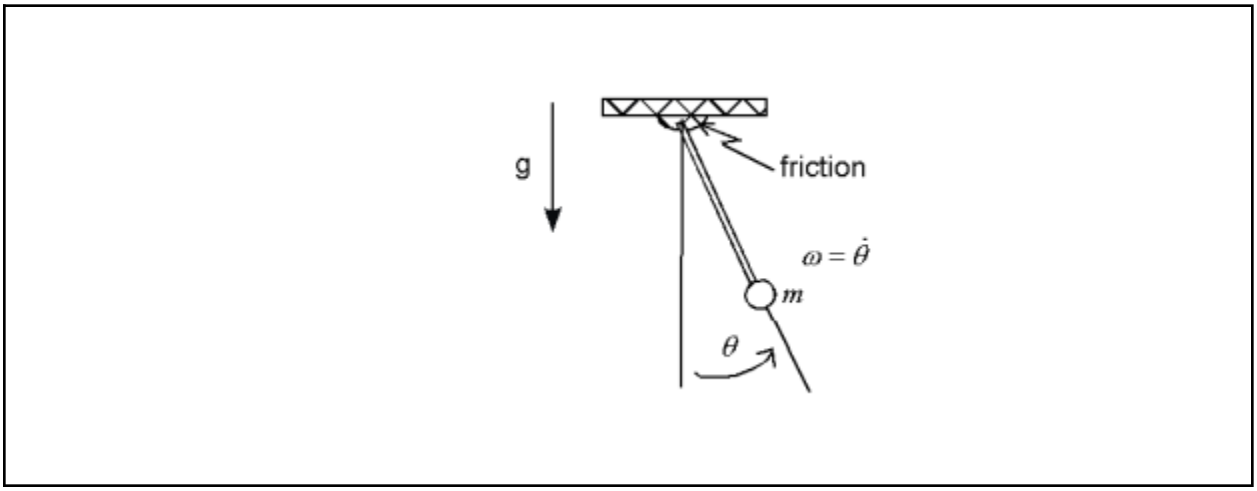

Figure 1: Physical model of simple damped pendulum

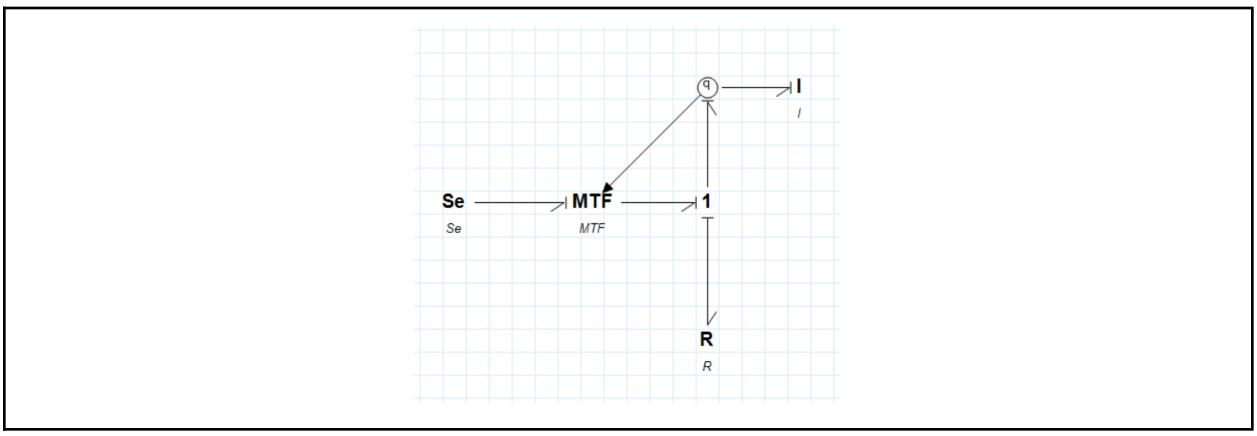

Figure 2: Bond graph model of simple damped pendulum

2. Analysis

A length of $l = 0.5m$, mass of $m = 0.8kg$, and friction torque of $d = 0.5Nm$ were used for simulation of the pendulum-clock. For the simple pendulum with friction, different friction

torques were also tested. The undamped natural frequency of the pendulum is shown in Eq.1, which was reflected in the simulation.

Eq1.

$$f_{natural} = \frac{1}{2\pi}\sqrt{\frac{g}{l}} = \frac{1}{2\pi}\sqrt{\frac{9.81}{0.5m}} = 0.705Hz$$

The signum function ('sign' in 20-sim) is used in the damping R element to simulate Coulomb friction.

```
parameters
   real d = 0.5;
equations
   p.e = d * sign(p.f);
```

Figure 3: 'R' block in bond graph models coulomb friction

The effort source ('Se' block) has a magnitude equivalent to the product of mass and gravity. This is scaled by $-\ lsin(r)$ by the 'MTF' block, since the torque on the pendulum due to gravity is opposite in sign of pendulum angle r. The pendulum angle r is provided as a signal from the 'QSensor' to the MTF block, and integrates the flow of the 1-Junction.

```
parameters
   real global l;
equations
   p2.e = -l*sin(r) * p1.e;
   p1.f = -l*sin(r) * p2.f;
```

Figure 4: 'MTF' block in bond graph models coulomb friction

The 'I' element represents the inertia of the pendulum, which is the product of mass and squared length, as the pendulum rod is assumed massless.

```
parameters
   real global m = 0.8;
   real global l = 0.5;
   real i = m*l^2;
equations
    state = int(p.e);
    p.f = state / i;
```

Figure 5: ‘I’ block in bond graph models inertia of pendulum

For the pendulum-clock, an effort source is added to the model which provides an impulse as the pendulum sweeps across the vertical (0 degree) position. The effort is applied in the direction of the angular velocity for the next $10\ \circ$ of angular motion. Figure 6 shows the pendulum-clock bond graph and figure 7 shows the ‘Se’ block logic for applying the impulses.

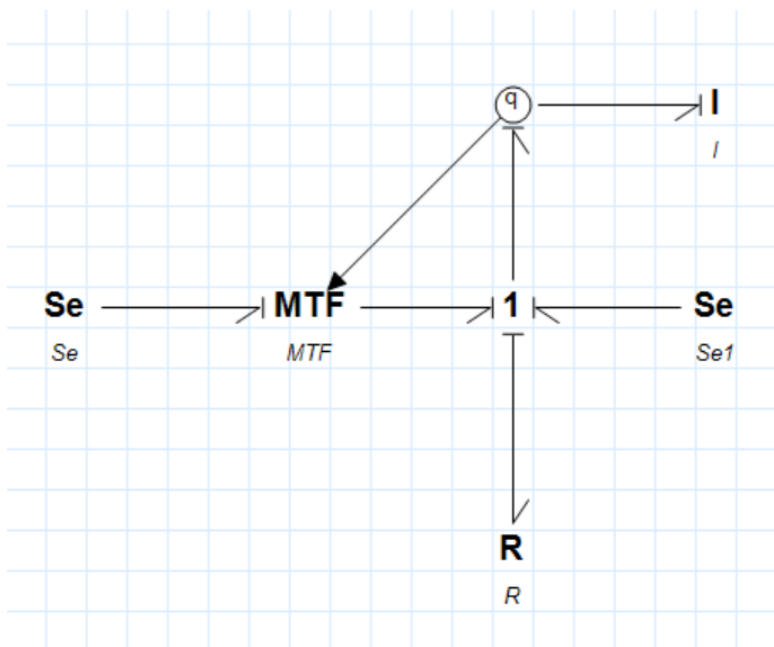

Figure 6: Pendulum-clock bond graph model

```
parameters
   real effort = 5;
variables
   real flow;
   real global angVeloc;
   real global angPos;
equations
   if (angVeloc>0 and angPos>0 and angPos<0.174533) then
      p.e = effort;
   elsif (angVeloc<0 and angPos<0 and angPos>-0.174533) then
      p.e = -effort;
   else
      p.e = 0;
   end;
   flow = p.f;
```

Figure 7: 'Se' block in pendulum-clock applies effort for 0-10◦ (0-0.174rad) of rotation

3. Results

Figure 8-11 below shows the simulation results for the simple damped swinging pendulum for different damping values, with initial conditions of $r_0 = \frac{\pi}{4} rad$ and $w_0 = 0 \frac{rad}{s}$.

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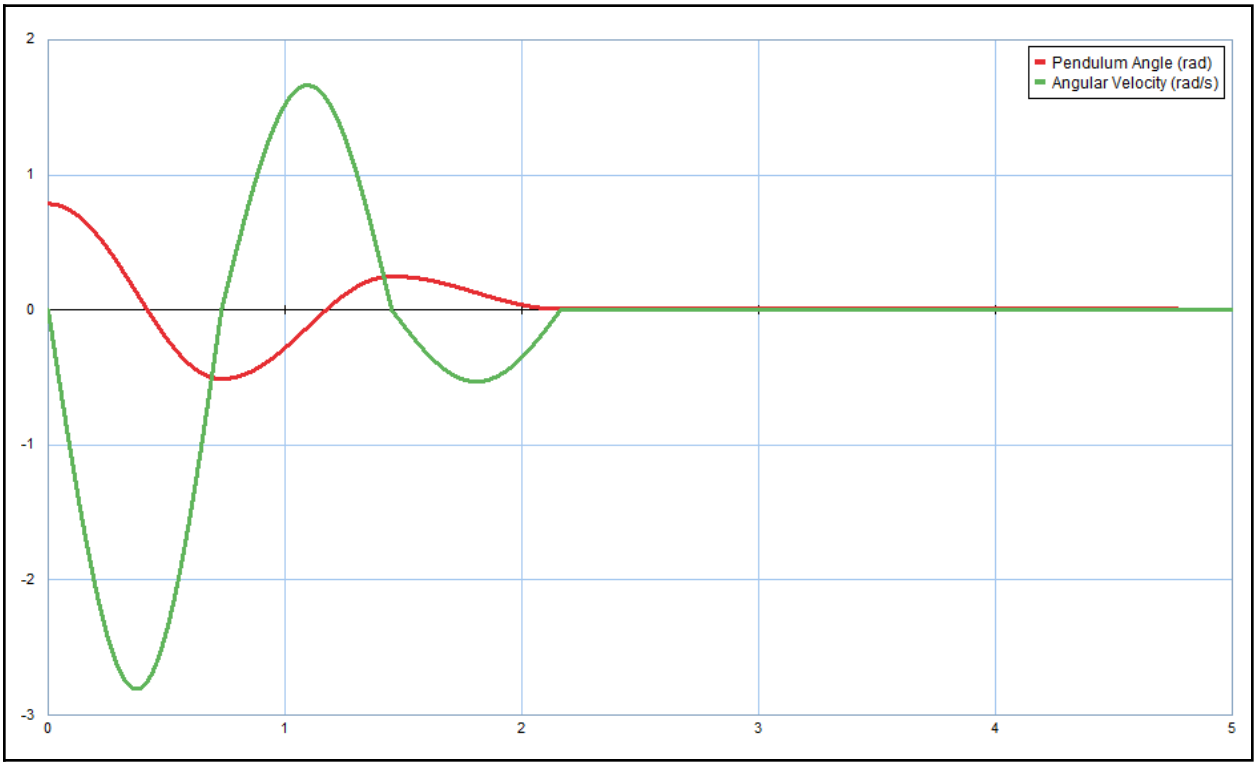

Figure 8: Simple damped pendulum simulation with R=0.5Nm

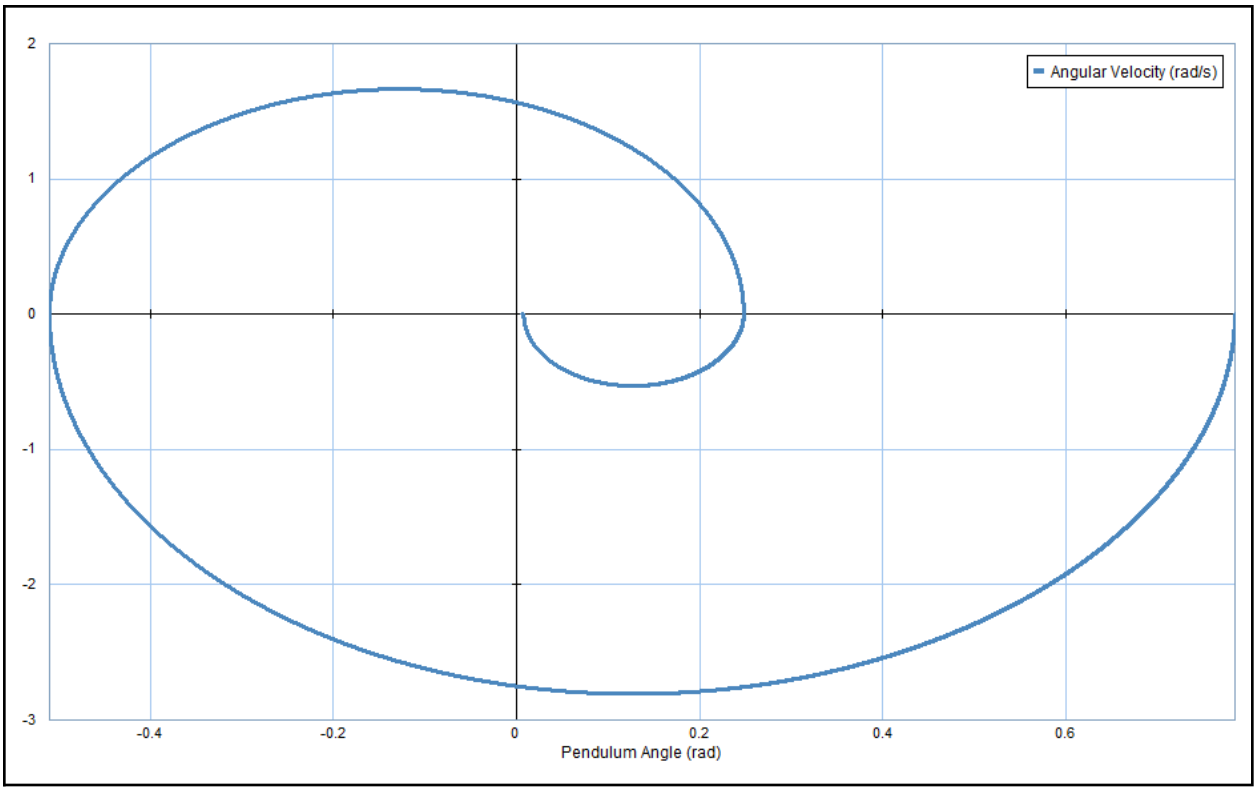

Figure 9: Simple damped pendulum simulation with R=0.5Nm (state-phase plot)

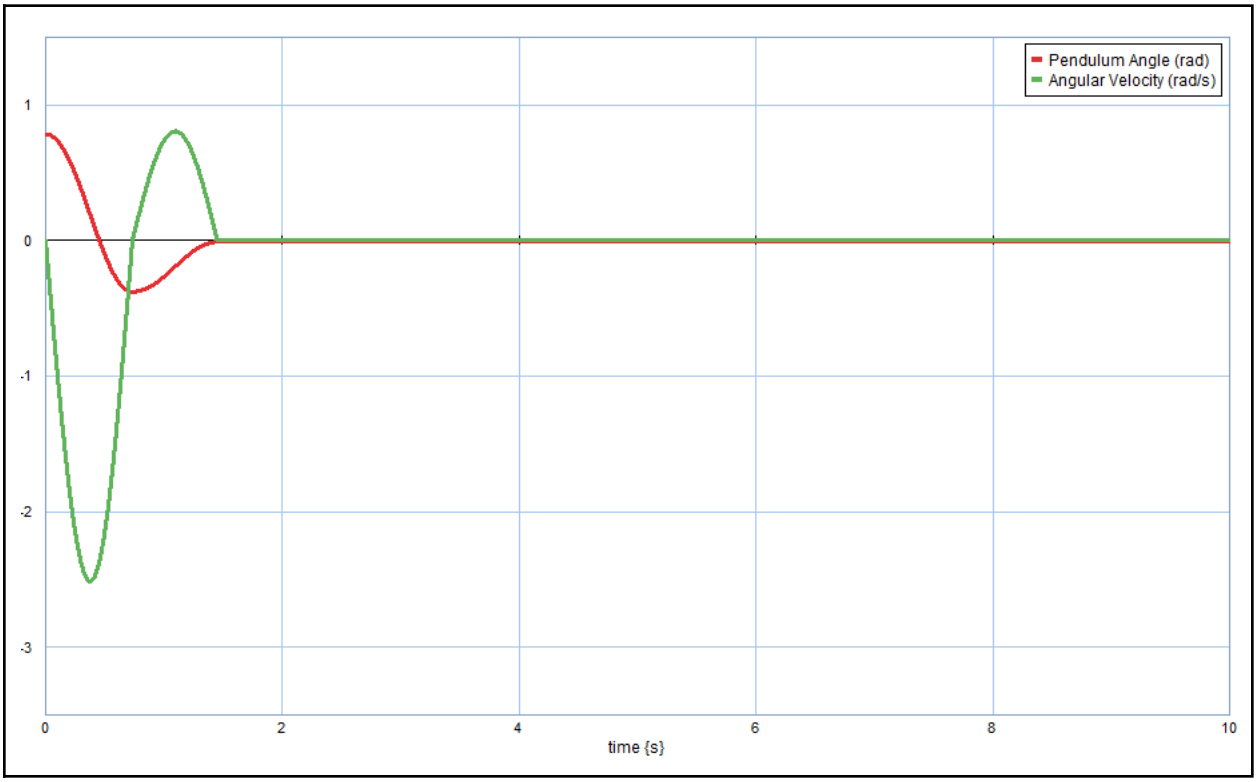

Figure 10: Simple damped pendulum simulation with R=0.75Nm

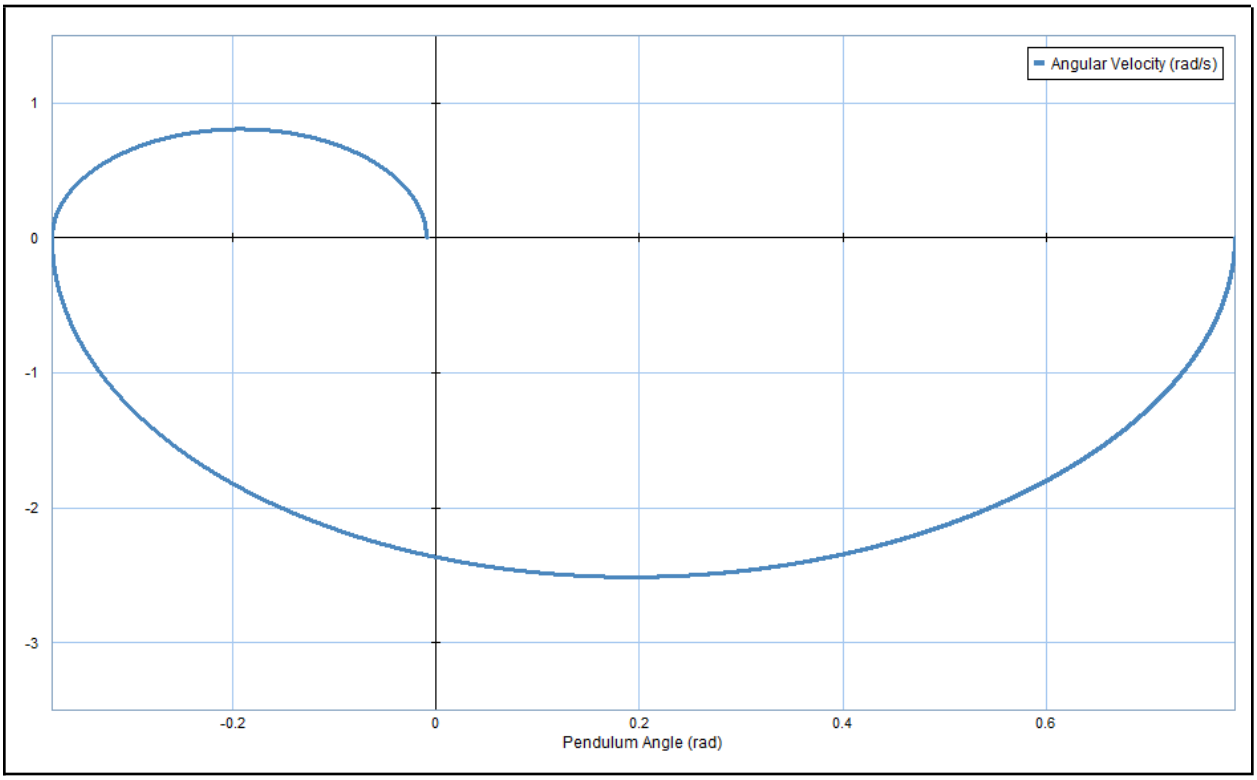

Figure 11: Simple damped pendulum simulation with R=0.75Nm (state-phase plot)

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Figures 12-19 below show the simulation results for the pendulum-clock with friction torque $d = 0.5Nm$ and initial conditions of $r_0 = \frac{\pi}{4} rad$ and $w_0 = 0\frac{rad}{s}$. A torque is applied to the pendulum between 0 and 0.174 radian when rotating positively, and between 0 and -0.174 radian when rotating negatively. In figures 12-13 a 5Nm torque is applied, in figures 14-15 a 2Nm torque is applied, in figures 16-17 a 1Nm torque is applied, and in figures 18-19 a 0.75Nm torque is applied.

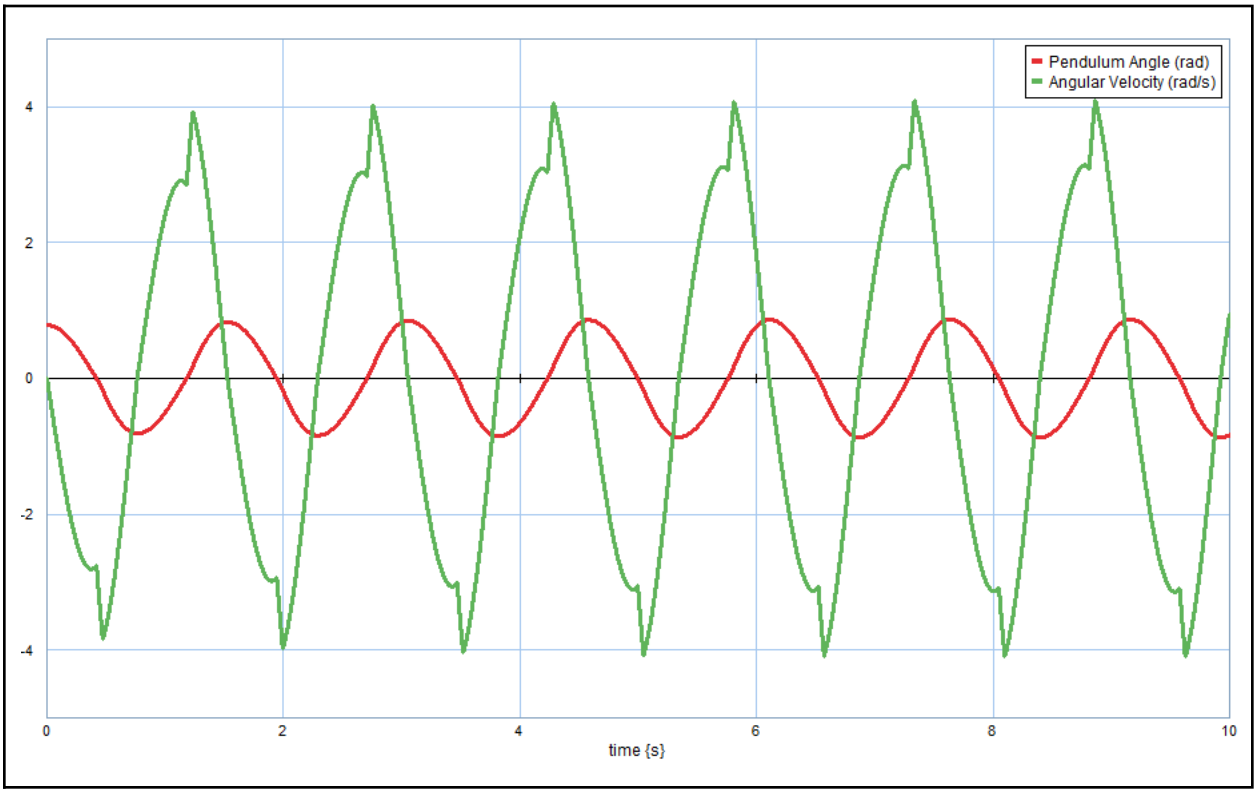

Figure 12: Pendulum-clock with 5Nm impulses applied for 0.174radians (~10◦)

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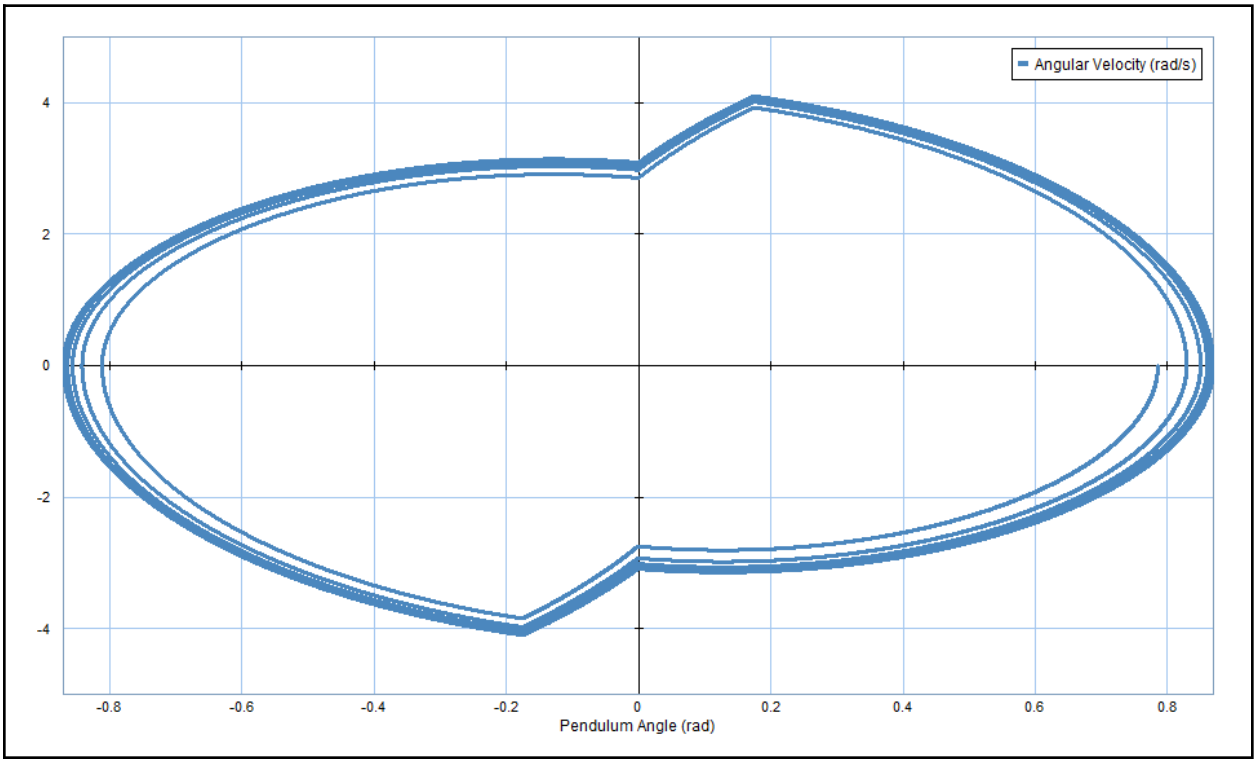

Figure 13: Pendulum-clock with 5Nm impulses applied for 0.174radians (state-phase plot)

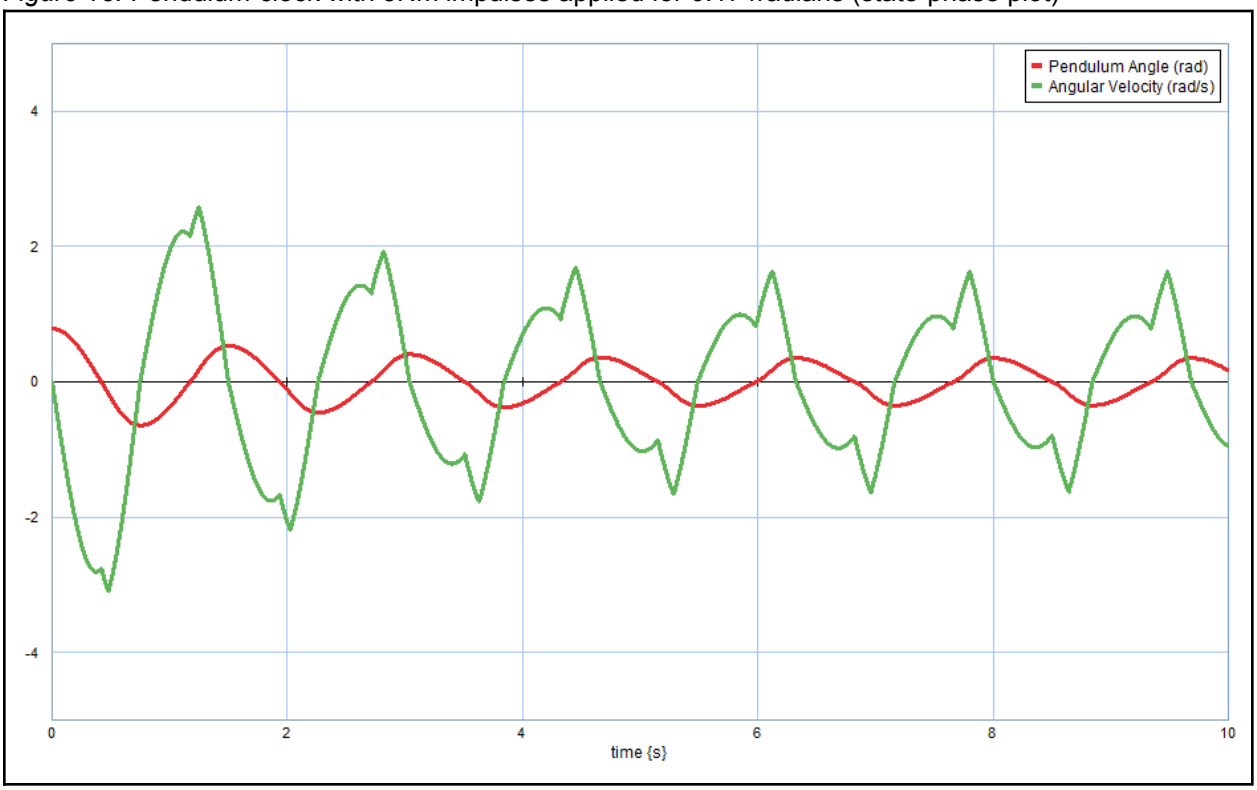

Figure 14: Pendulum-clock with 2Nm impulses applied for 0.174radians (~10◦)

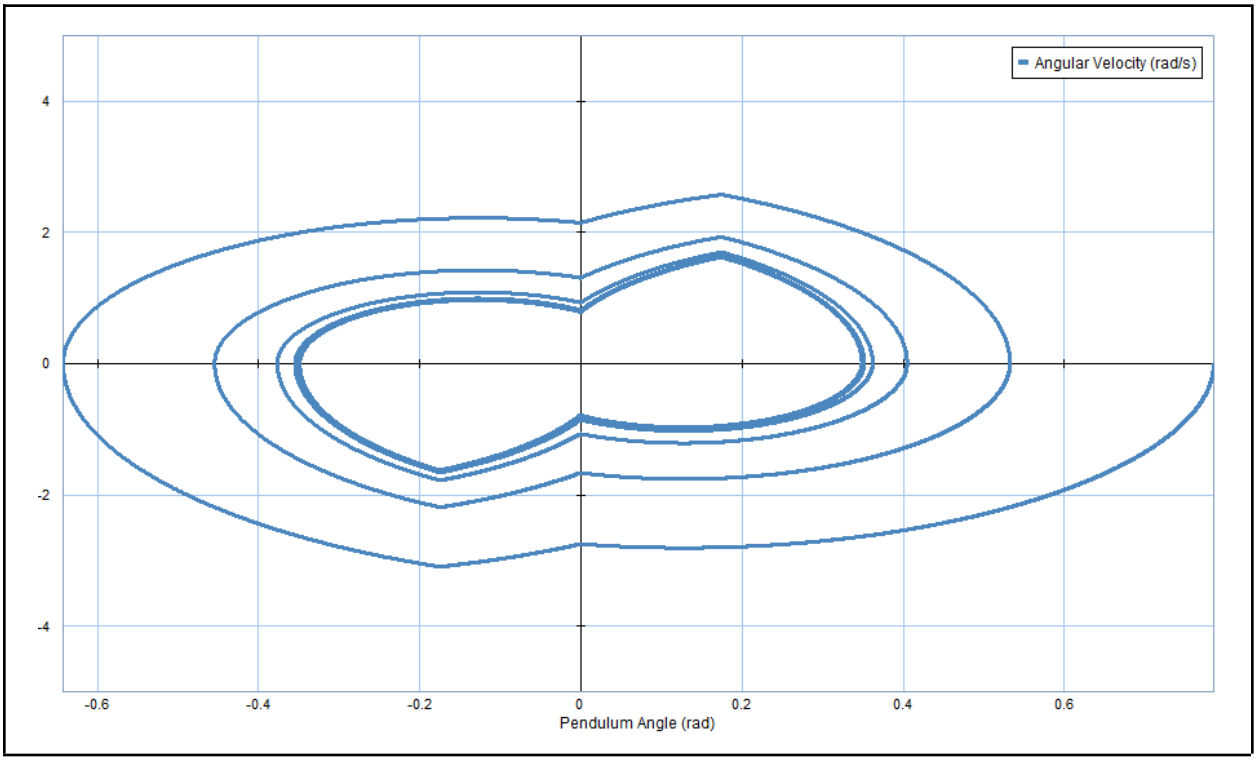

Figure 15: Pendulum-clock with 2Nm impulses applied for 0.174radians (state-phase plot)

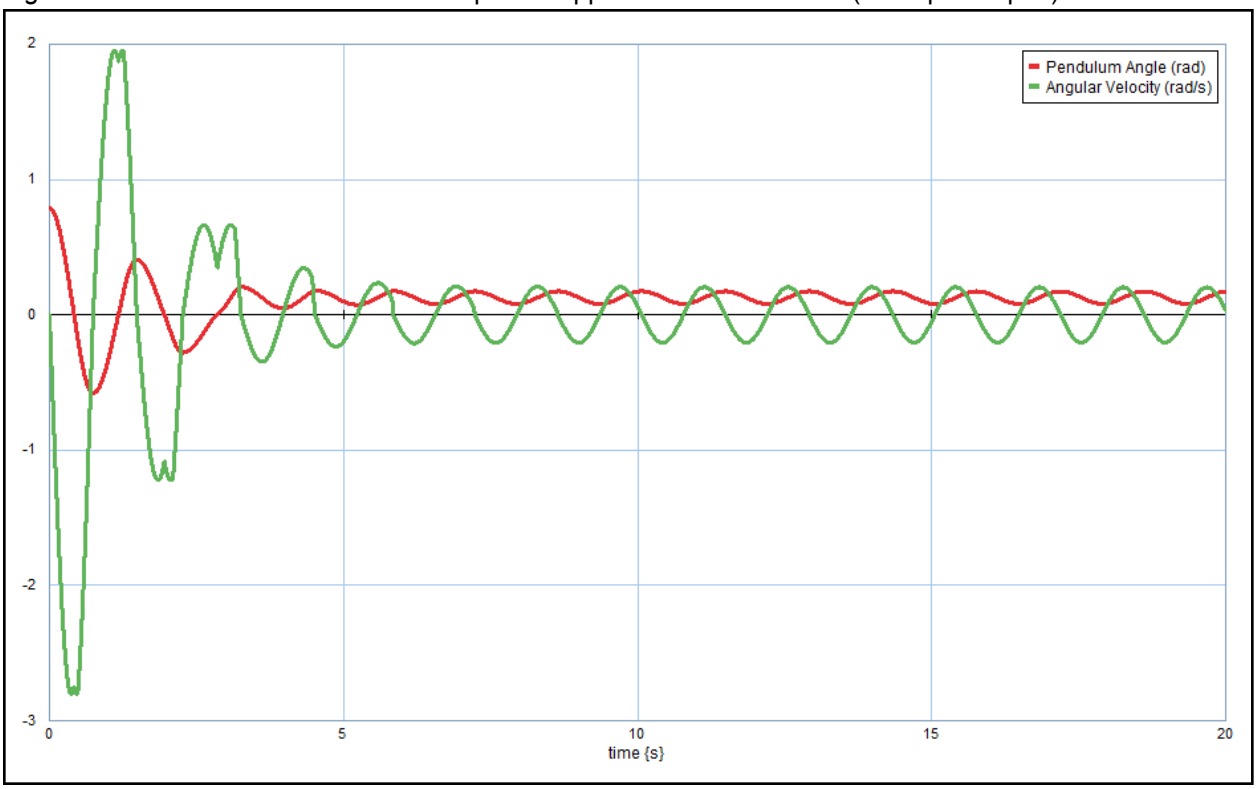

Figure 16: Pendulum-clock with 1Nm impulses applied for 0.174radians (~10◦)

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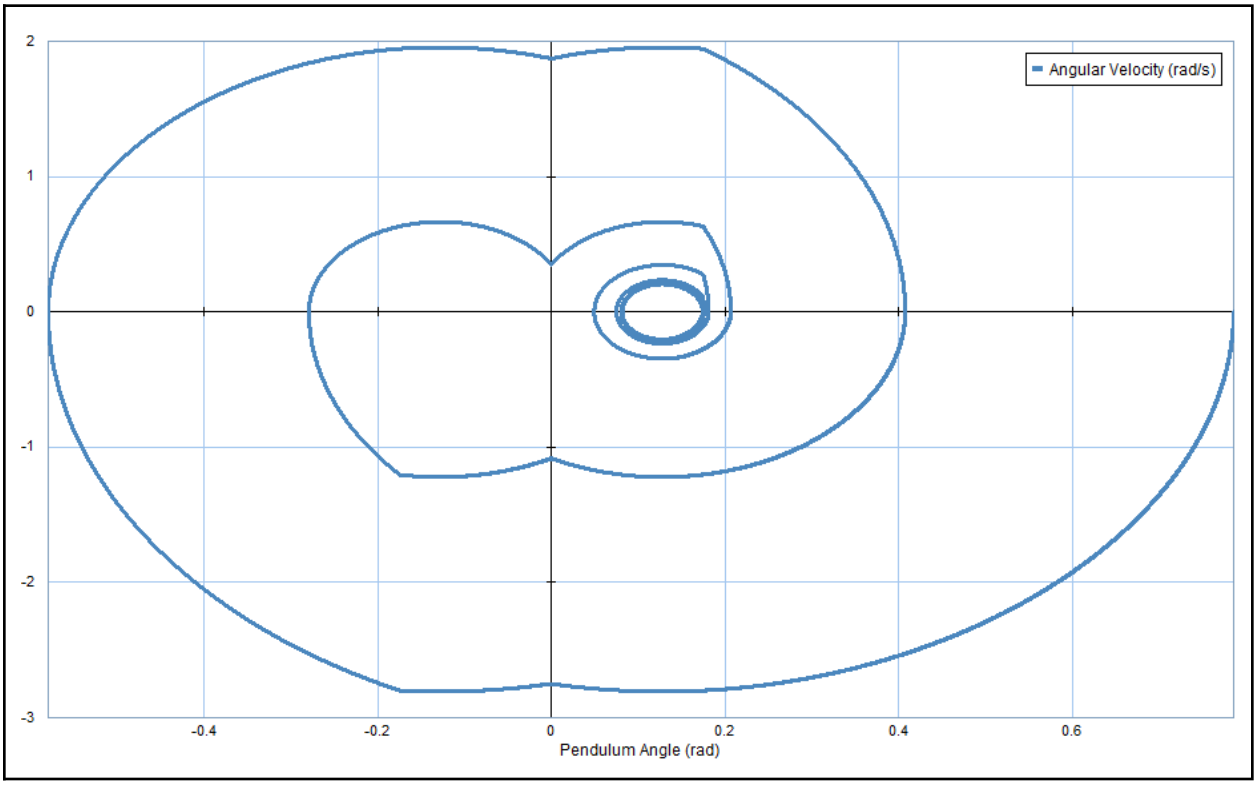

Figure 17: Pendulum-clock with 1Nm impulses applied for 0.174radians (state-phase plot)

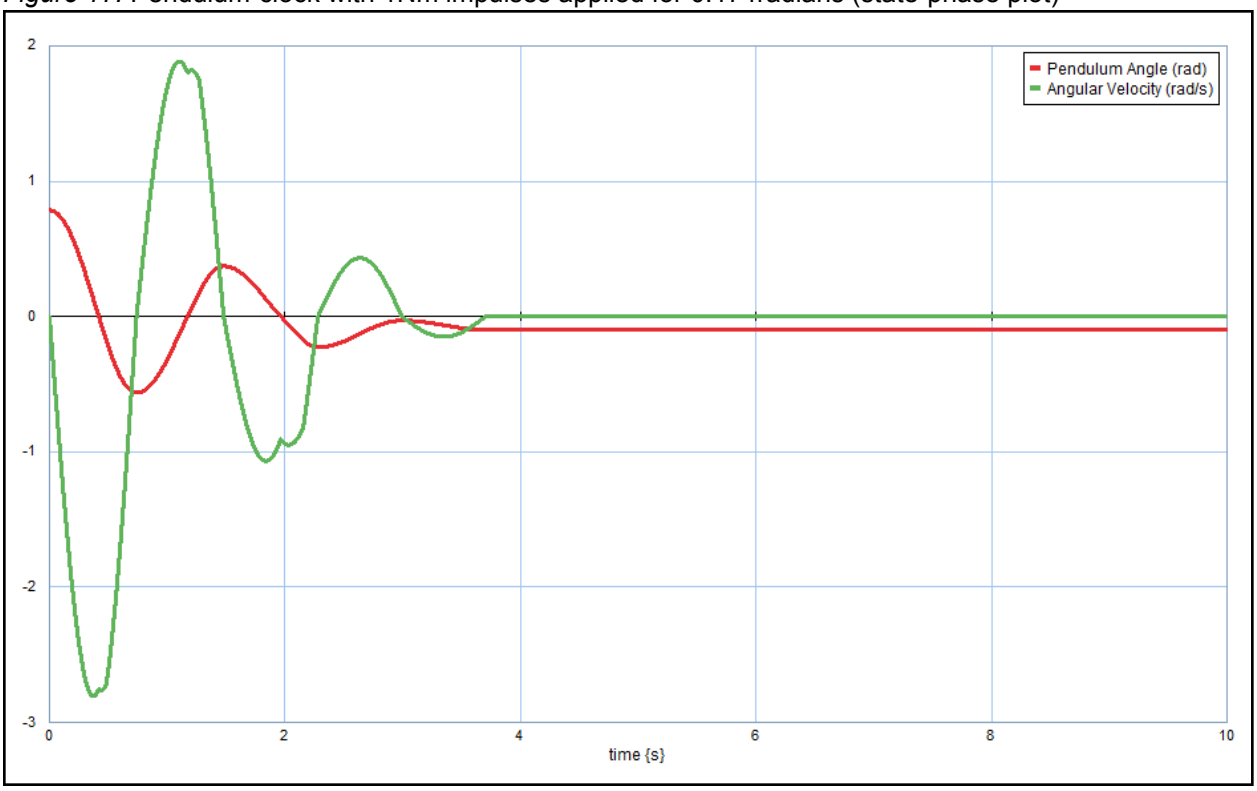

Figure 18: Pendulum-clock with 0.75Nm impulses applied for 0.174radians (~10◦)

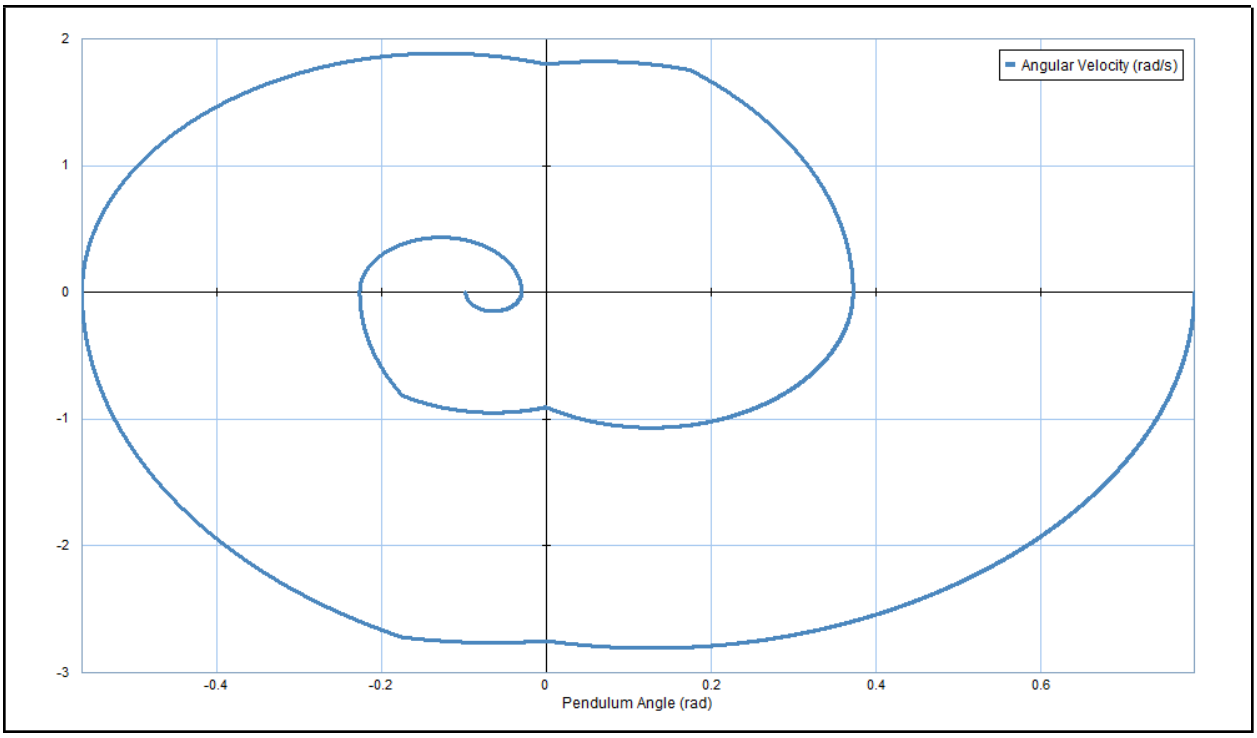

Figure 19: Pendulum-clock with 0.75Nm impulses applied for 0.174radians (state-phase plot)

4. Discussion

Figure 8 and 9 (simple pendulum with 0.5Nm friction torque) show an underdamped system oscillating at its natural frequency. As expected in figure 10-11, increasing the friction torque to 0.75Nm reduces the number of free oscillations.

Figure 12-13 shows that applying a 5Nm torque for 10 degrees of rotation twice a cycle is a large enough impulse to achieve a limit cycle, and the pendulum continues to oscillate at approximately $\pm \frac{\pi}{4} rad$ $(\pm\ 45\ \circ)$. However, with 0.5Nm friction torque damping the pendulum, only a much smaller impulse is needed to achieve a limit cycle. Figure 14-15 shows that a limit cycle is achieved by applying 2Nm for 10 degrees of rotation twice a cycle, and results in steady state oscillations of $\pm\ 0.35 rad$ $(\pm\ 20\ \circ)$. A 1Nm torque scarcely achieves continued oscillations, which are no longer symmetric (figure 16-17). Further reducing to 0.75Nm causes the oscillations to die off within 3 oscillations (figure 18-19).

5. Conclusion

A damped pendulum was modeled in 20-Sim using a bond graph and different impulse strengths were tested to achieve a limit cycle (sustained oscillations). For our tests where impulses were applied twice per oscillation for 10◦ of rotation, 2Nm impulses were the smallest that were able to achieve a symmetric limit cycle (oscillates continuously between positive and

negative angles). For a real system, ~2Nm impulses would likely be reliable while keeping the oscillations reasonably small ($\leq 20 \circ$ each side).